PHARMACOECONOMICS





Lecture 4

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PHARMACOECONOMIC METHODS



Economic

Cost consequence
Cost benefit
Cost effectiveness
Cost minimization
Cost utility

Humanistic

Quality of life
Patient preferences
Patient satisfaction

Cost Benefit Analysis (CBA):

- Measures the value of all resources consumed (costs) and benefits (outcomes) in **monetary terms**.

- Both the costs and the benefits are measured and converted into equivalent dollars ($), in the year in which they will occur, **thus alternatives with completely different outcomes can be compared.**

- **An advantage** of this type of analysis is that many **different outcomes can be compared** as long as the outcomes measures are valued in monetary units.

- **The disadvantage** is that placing economic values on medical outcomes **is not an easy task** and there is **no universal agreement** on one standard method for performing this.

Basic idea of Cost-Benefit Analysis (CBA)

- Benefits of a program: +$$$
- Costs of a program: - $$$

- Net benefits: ? $$$

- Ideally, all costs and benefits resulting from the program should be included. This presents considerable difficulty, especially on the benefits side of the equation, as many benefits are either difficult to measure, difficult to convert to dollars, or both.

- Many CBAs measure and quantify **direct costs and direct benefits** only due to difficulties in measuring indirect and intangible costs and benefits.

- For example, the benefits of improved patient quality of life, patient satisfaction with the healthcare system, and working conditions for the physician are not only difficult to measure, but are extremely difficult to assign a monetary (a dollar) value to them.

- This approach is **not widely** used in health economics.

- To illustrate the advantage of CBA compared with CEA, the following table shows examples of various programs and interventions and their corresponding cost-effectiveness and cost-benefit ratios. **(CBA ratios are expressed as benefit-to-cost ratios, where the higher the number, the more cost-beneficial)**

Comparison of Cost-Effectiveness Ratios and Benefit-to-Cost Ratios

Program or Intervention:	Cost-Effectiveness Ratio	Benefit-to-Cost Ratio
AIDS prevention and awareness program	$230,000/case prevented	8.4:1
Vaccination program for children	$104,000/case prevented	0.3:1
Smoking cessation intervention	$3,700/quit	6.7:1
Diabetes medication adherence program	$67/normoglycemic patient	15.1:1
Breast cancer screening program	$50,000/life year saved	2.4:1

- Assume you are a decision maker and you must choose one program from the table to implement in your organization.

- If you only had cost-effectiveness ratios available to help make the choice. How would you choose? One can quickly see that it would be difficult to compare the programs using only cost-effectiveness ratios because of ?????????????

➢ because of the varying outcomes (e.g., case prevented, life years saved, normoglycemic level).

- **On the other hand, the benefit-to-cost ratios can be ranked, and programs with similar, as well as dissimilar, outcomes can be compared.**

- If the goal of the decision maker is to maximize the investment, **the program with the highest benefit-to-cost ratio** (in this case, diabetes medication adherence program) would be chosen. If only the cost-effectiveness ratios were available, it would be more difficult to compare the value of the various interventions.

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- Pharmacists are working in many areas of health care providing clinical services and developing specialty clinics in areas such as anticoagulation, diabetes, asthma, osteoporosis and HIV.

- Although clinical pharmacy programs have shown improvements in clinical outcomes such as glycemic control, bone density (osteoporosis) and asthma control, financial pressures are forcing decision makers to consider the following questions:

  - Do the benefits (outcomes) of a program or intervention outweigh (i.e. exceed) the costs?
  - Which program will provide the greatest benefit (outcome)?

- CBA is a tool that can be used to address (i.e. answer) these questions.

- The unique aspect of placing a monetary value on the outcome or benefit in CBA also **presents a challenge or disadvantage of the method.**

- For example, when comparing the cost-effectiveness ratios for an AIDS prevention and awareness program with the vaccination program for children, it would appear that the vaccination program would be the most cost-effective. But, when examining the benefit-to-cost ratios, the AIDS program is more cost-beneficial.

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- In this example, **the benefit** (case prevented) **was valued higher for AIDS patients** than vaccinations for children.

- The controversy arises regarding the methodology used to place a dollar value on a case prevented or a human life saved.

- **Thus, the advantage of placing a dollar value on the benefit or outcome is also a disadvantage of the method.**

CBA results can be presented in the following formats:

1. Calculate the **Benefit-to-Cost (or Cost-to-Benefit) ratio**

- CBA results can also be calculated by summing up the total benefits and dividing by the total costs.

- The ratio may be expressed as:

  - a benefit-to-cost ratio (Benefit ($)/Cost ($))
  - or a cost-to-benefit ratio (Cost ($)/Benefit ($)).

- Depending on how the ratio is calculated, interventions are **cost-beneficial** if: Benefit-to-cost > 1 or Cost-to-benefit < 1

2. Or calculate the **Net Benefit (or Net Cost)**

- Simply presents the difference between the total costs and benefits.

Net benefit= total benefits ($) - total costs ($)

Net cost= total costs ($) - total benefits ($)

- Intervention would be considered to be **cost-beneficial**

  if: Net Benefit > 0 or Net Cost < 0

Cost benefit analysis (CBA)

When to use	For resource allocation decisions – ascertain whether the beneficial outcomes justify the cost. Comparison of different programs with different outcomes.
Application	Scope of analysis is usually broad, addressing large societal issues. • AIDS prevention and awareness programs • Smoking cessation intervention • Diabetes drug adherence • Breast cancer screening
Cost unit: Outcomes unit: Calculation	Monetary value Measured in similar/different units and are always valued in monetary units (e.g. amount willing to pay to prevent a death, amount willing to pay to reduce exposure to a hazard) Benefit to Cost ratio = Benefit ($)/Cost ($) Net Benefit = Benefit ($) – Cost ($)
Interpretation	Choose alternative with the largest value (i.e. benefit)

Cost Utility Analysis (CUA):

- Is an economic tool (i.e. evaluation) in which the intervention consequence is measured in terms of **quantity and quality of life**.

- It is much the same as cost-effectiveness analysis. Some consider **cost-utility analysis (CUA) a subset of CEA** because the outcomes are assessed using a special type of clinical outcome measure, usually the quality-adjusted life-year (QALY).

- The CUA takes patient preferences, also referred to as **utilities**, into account when measuring health consequences.
- As mentioned above, the most common outcome unit used in CUA is the QALY, which incorporates both the quality (morbidity) and quantity (mortality) of life.

Utility: the state of being useful or beneficial.

- Other outcome units that are seen less frequently include disability-adjusted-life years **(DALYs)** and healthy-year equivalents **(HYEs).**

- The advantage of a CUA is that **different types of health outcomes and diseases with multiple outcomes** of interest can be compared (unlike in CEA) using one common unit such as the QALY.

- CUA incorporates morbidity and mortality into this one common unit without having to determine or estimate the monetary value of these health outcomes (unlike CBA).

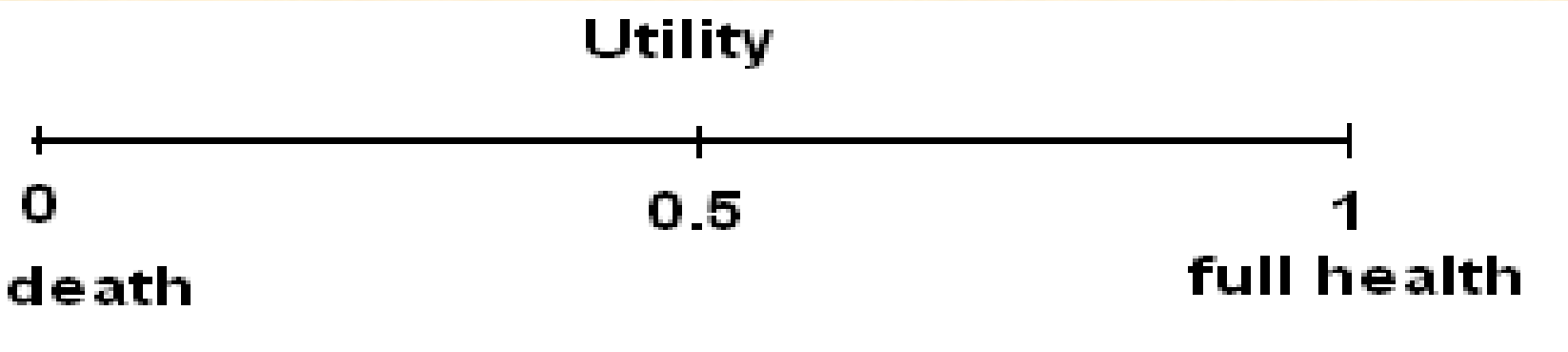
- **The disadvantage** of this method is that it is difficult to determine an accurate utility. Therefore, although the number of CUA research articles in the literature is increasing yearly, the methods for estimating utilities/QALYs may not be fully understood.

- HRQOL measure is an utility, having value between 1.0 (perfect health) and 0.0 (death).

- To estimate **utility weight (score)** for various conditions two broad methods are used to generate, these scores: **direct elicitation** methods (rating scale, standard gamble, and time tradeoff) and **indirect elicitation** methods, using standardized weightings (e.g., EQ-5D and SF-6D surveys).

- Quality-adjusted life years (QALYs) are then calculated by multiplying the **time** in a health state by the appropriate **utility score**.

Quality Adjusted Life Years



- QALY = Utility x # years in a health state
- One year at full health QALY = 1.0
- Death QALY = 0.0
- 3 years disabled (U =0.5) = 1.5 QALYs

- This method is well suited to the **evaluation of chronic diseases that have deleterious effects on HRQOL.**
- Differences between treatments are expressed as the **incremental cost per QALY gained.**
- CUA can compare cost, quality, and the quantity of patient-years.

- CUA is employed less frequently than other economic evaluation methods because of a lack of agreement on measuring utilities, difficulty comparing QALYs across patients and populations, and difficulty quantifying patient preferences.

- In CEA, the costs are measured in money and there is a defined outcome. But in CUA, the outcome is an unit of utility (e.g. a QALY).

- **Intermediate outcomes** are **not appropriate** to use in cost-utility analysis

- Results of CUA are expressed in a ratio, a cost-utility ratio (C:U ratio).

Calculating QALYs (Example)

With treatment X

Estimated survival = 10 years
Estimated quality of life
(relative to 'perfect health') = 0.7
QALYs = (10 X 0.7) = 7.0

Without treatment X

Estimated survival = 5 years
Estimated quality of life
(relative to 'perfect health') = 0.5
QALYs = (5 X 0.5) = 2.5

QALY gain from treatment X = 7 - 2.5 = 4.5 QALYs

If the cost of treatment X is £18,000 then the cost per QALY is £4,000 per QALY
(£18,000 divided between 4.5 additional QALY's)

- we compare two treatment options, drug A and drug B. Although drug B extends the person's life for more years, the quality of life for those years (for B) is lower than with drug A.

	Cost for Treatment (dollars)	Years of Life Saved	Utility for Each Year of Life Saved	QALYs
Drug A	$10,000	**5**	**0.8**	**4.0**
Drug B	$20,000	**7**	**0.5**	**3.5**
Calculation			Result	
CEA	($20,000 − $10,000) / (7 years – 5 years)		$5,000 per extra year of life	
CUA	($20,000 − $10,000) / (3.5 QALYs – 4.0 QALYs)		Drug A dominant	

- If a CEA were conducted, option B would be relatively cost-effective at an incremental cost per year of life of $5,000.

- If the quality of those years is incorporated into the equation by calculating QALYs, option A becomes dominant in that it costs less and provides a better outcome (more QALYs).

Cost utility analysis (CUA)

When to use	Similar to CEA with the added dimension of a particular point of view, most often the patient. Intervention outcomes /consequence is measured in terms of patient preference (e.g. quantity and quality of life).
Application	CUA should be used when QoL is the important outcome: • Psychological well-being, physical and social function are important in the treatment of arthritis • Chemotherapy may increase survival but decrease well-being, both quality and quantity of life are important
Cost unit: Outcomes unit: Calculation	Monetary value Health benefits across therapies are valued in similar units based on individual preferences Utility values (QALY = years in health state *utility) Cost per QALY gained or Cost per HYE gained
Interpretation	Lowest cost per unit QALY gained

Summary of costs & outcomes for methods

Method	Cost Unit	Outcomes measure	Interpretation of result
CMA	Monetary	Natural units (equivalent)	Choose option with lowest cost
CEA	Monetary	Natural clinical units (life-years gained, mmol/l blood glucose, mm Hg blood pressure)	Lowest cost per unit of effectiveness Incremental cost
CUA	Monetary	Quality adjusted life years (QALYs) or other utilities	Lowest cost per unit QALY
CBA	Monetary	Monetary	Ratios of greater than 1.0
COI	Monetary	Monetary	Burden of disease / budget impact

Questions that pharmacoeconomics may help to address are as follows:

- What drugs should be included on the hospital formulary?
- What is the best drug for a particular patient?
- What is the best drug for a pharmaceutical manufacturer to develop?
- Which drug delivery system is the best for the hospital?
- How do two clinical pharmacy services compare?
- Which drugs should be included in a Medicaid formulary?
- What is the cost per quality-adjusted year of life extended by a drug?
- Will patient quality of life be improved by a particular drug therapy decision?
- What is the best drug for this particular disease?
- What are the patient outcomes of various treatment modalities?

The various applications of pharmacoeconomics include:

- Clinical decisions for individual patient treatment-(CEA & CUA)
- Justification/evaluation of clinical services/programs – (CBA)
- Formulary decision making – (CEA & CUA)
- To support drug policy - (CEA & CUA)
- Resource allocation - (CEA , CUA & CBA)

Importance of Pharmacoeconomics:

- Pharmacoeconomic analysis helps to achieve maximum benefit in limited cost.
- Clinicians want their patients to receive best care and outcome available.
- The payers want to manage rising costs.
- Pharmacoeconomics combines the objectives of both clinician and payers by estimating the value of patient outcomes for the expenditure spent on medications and other healthcare services.
- In today's healthcare settings, pharmacoeconomic methods can be applied for effective formulary management, individual patient treatment, medication policy determination, and resource allocation.


THANK
YOU

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